

Eight-year ecophysiology and growth dynamics of *Picea rubens* seedlings planted in harvest gaps of partially cut stands

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ABSTRACT

In order to compensate for regeneration failure of red spruce (*Picea rubens*) in the managed temperate mixedwood forest, an enrichment planting study was conducted in yellow birch (*Betula alleghaniensis*)–conifer stands, in Quebec, Canada. The experimental design compared four cutting treatments: an uncut control (26 m²·ha⁻¹ of merchantable basal area [BA], with mean natural canopy gap areas of ≤ 100 m² each) and three hybrid single-tree and group-selection cuttings corresponding to a light cut (20% removal; residual BA [RBA]: 21 m²·ha⁻¹; mean harvest gap area: 220 m²), a moderate cut (31% removal; RBA: 18 m²·ha⁻¹; gap area: 270 m²), and a heavy cut (42% removal; RBA: 15 m²·ha⁻¹; gap area: 340 m²). In the cuts, large-sized containerized seedlings of red spruce were planted within harvest gaps, while in the control, seedlings were planted systematically, for comparison. Three years later, a mechanical release treatment was applied in half of planted plots to limit competing vegetation. After eight years, seedling survival in the three cuts was equally high (> 80%), and greater than in the control (40%). Morphophysiological performance of seedlings was best in the heavy cut creating the largest gaps. In general, the release treatment enhanced survival in the control and in the light cut with the smallest gap sizes. Release also improved morphophysiological performance (which, however, remained suboptimal) in all cutting treatments, but especially in the heavy cut. Although red spruce is highly shade-tolerant, the low survival and performance in the control suggest that for this species, seedlings grown with traditional forest nursery practices are not adapted to plantation under a closed canopy. Results indicate that enrichment planting in intermediate-size harvest gaps (of about 340 m² in area) of partially cut stands (of about 40% in BA) combined to mechanical release could be a good strategy to maintain or reintroduce red spruce and promote its morphophysiological performance.

1. Introduction

Over time, the historical and progressive decline of certain companion tree species in managed forests can become an important biodiversity issue. In the context of global change, conserving these species contributes to ecosystem resilience (Lindenmayer and Franklin, 2002). Because each companion species generally represents a low proportion of total stand basal area, past silvicultural practices were not focused on providing the conditions required for their successful regeneration. This is the case of red spruce (*Picea rubens* Sargent), a keystone and desired commercial tree species of the temperate mixedwood forest of north-eastern North America (Gordon, 1996, Dumais and Prévost, 2007). The reproductive strategy of this highly shade-tolerant species is to establish a bank of seedlings and saplings over time under the forest canopy (Blum, 1990, Seymour, 1995, Wu et al., 1999), also called “advance regeneration” (e.g., Ferguson, 1984). When this advance regeneration is

scarce, as in many managed mixedwood stands (e.g., Prévost et al., 2010, Raymond et al., 2018), natural establishment of new red spruce seedlings through silviculture appears as a lengthy and complicated process that depends on many ecological factors (Dumais and Prévost, 2016). For example, since the species’ roots are slow-growing and fine (Klein et al., 1991, Greenwood et al., 2008), young seedlings show slow initial growth, are vulnerable to crushing by hardwood litter and have a high mortality rate (Dumais and Prévost, 2016).

Silvicultural approaches based on small-scale plantation have been proposed to maintain or reintroduce red spruce, and accelerate its succession in managed mixedwood stands in which regeneration is deficient or absent (Dibble et al., 1999, Prévost et al., 2003, Dumais and Prévost, 2016). Recently, experimental fill planting was reported to increase red spruce stocking in a cutover mixedwood stand in Maine, USA (Kenefic et al., 2014). In Quebec, Canada, enrichment planting under different forest canopy densities and types is currently under

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investigation (Dumais et al., 2019). In general, red spruce seedlings and saplings benefit from the presence of a partial canopy that moderates the microclimate by preventing extreme light, temperature and humidity fluctuations (Vann et al., 1994, Alexander et al., 1995, Day, 2000, Major et al., 2003), while limiting the development of competing species (Dibble et al., 1999, Raymond and Bédard, 2017, Dumais et al., 2019). However, even if recent results suggest that red spruce could initially perform well when reintroduced by enrichment planting under partial forest covers, information is lacking about longer term trends and the need for tending treatments (Dumais et al., 2019). Nevertheless, observations made on small (< 1.3 m in height) natural seedlings present as advance regeneration confirmed that uniform partial canopy removal could not provide sufficient light to optimize physiological and growth performances in the long term, despite the initial advantage provided by the residual canopy for survival and acclimation (Dumais and Prévost, 2008). In contrast, partial cuts that incorporated small to intermediate-size gaps (100–300 m² in area) created microenvironmental conditions promoting simultaneously survival, acclimation, and competitive growth of small advance red spruces (Dumais and Prévost, 2014). Consequently, enrichment planting in that range of gap sizes appears as an interesting option to maintain or reintroduce the species efficiently in managed mixedwood stands.

Ecophysiological performance of seedlings (Burdett, 1990, Margolis and Brand, 1990, Grossnickle, 2000) in harvest gaps is influenced by gap characteristics (e.g., size, shape, slope and aspect) and seasonal and daily light, temperature and humidity conditions (Canham, 1988, Weiss, 2000, Prévost and Raymond, 2012, Kern et al., 2017). Few studies have examined acclimation and development of planted conifer seedlings in harvest gaps and most of them were conducted in much larger gap sizes (i.e. patches). In British Columbia, Canada, de Montigny and Smith (2017) reported that opening size should be at least 2400–3300 m² for adequate height growth of planted Douglas-fir (*Pseudotsuga menziesii* (Mirbel) Franco). In Quebec, Canada, white spruce (*Picea glauca* Moench Voss) planted in patch clearcuts ranging from 500 to 10 000 m² had the best height growth in the smallest openings, likely because of the edge effect that tempered the microclimate and decreased interspecific competition (Hébert et al., 2013). In contrast, York et al. (2004) reported a marked reduction in planted seedling height growth along the edges within the openings for six native conifers in California, USA. To our knowledge, no study has looked at the ecophysiology and growth of planted red spruce seedlings in harvest gaps. Considering the costs of seedling production and planting operations, understanding the factors and mechanisms driving the success of red spruce enrichment planting is important before it is deployed at an operational scale.

The present study tests the red spruce enrichment planting in managed yellow birch–conifer stands. Our aim was to characterize morphophysiology, acclimation, and growth of these seedlings that were planted under intact forest canopy (uncut control) and in intermediate-size harvest gaps created by a gradient of hybrid single-tree and group-selection cutting intensity (Raymond et al., 2018). This silvicultural system closely emulates the natural canopy gap disturbances occurring in the temperate mixedwood forest (Kneeshaw and Prévost, 2007, Duchesne and Prévost, 2013). It can also promote natural regeneration of both red spruce and yellow birch (Dumais and Prévost, 2014, Prévost and Charette, 2015, Raymond et al., 2018). To limit the impact of competing vegetation, we also included a mechanical release treatment in the experiment. In this paper, we present the eight-year results pertaining to ecophysiology and growth dynamics. Building on the conclusions of our previous work, we verified three predictions: survival and morphophysiological performance of planted red spruce seedlings will be (i) much greater in cuts (harvest gaps) than in the uncut control, despite a certain degree of acclimation in the latter (Dumais and Prévost, 2008, 2014), (ii) greater but not optimal in the heavy cut (the largest gaps, averaging 340 m² in area), because of

limited light availability (Dumais et al., 2019), and (iii) greater in released plots than in unreleased plots, because of the well-known negative impact of abundant non-commercial competing vegetation on planted spruce seedlings in yellow birch–conifer stands (e.g., Dumais and Prévost, 2019).

2. Materials and methods

2.1. Experimental site and design

The study was conducted in late-successional uneven-aged yellow birch–conifer stands located approximately 80 km northwest of Québec City, in the Portneuf County, QC, Canada (lat. 46° 59' N, long. 71° 59' W). The site is characterized by steep and hilly topography with rounded summits (altitude: 300–450 m) and podzolic soils developed from deep glacial till. These acidic soils are generally characterized by organic surface horizons and underlying horizons that are enriched with amorphous material composed of humified organic matter, aluminum and iron (Soil Classification Working Group, 1998). Mean monthly temperatures vary from −14.6 °C in January to 17.3 °C in July, and annual precipitations average 1250 mm (~30% as snow).

Before the cut, stand merchantable basal area (BA, all trees with diameter at breast height [DBH, measured 1.3 m above the ground] ≥ 9 cm) of 25.7 m²·ha^{−1}, composed of 42% yellow birch, 20% red spruce, 16% balsam fir (*Abies balsamea* (Linné) Miller), 12% sugar maple (*Acer saccharum* Marshall), 6% red maple (*Acer rubrum* Linné), and 4% hardwood companion species such as paper birch (*Betula papyrifera* Marshall). Mean height of dominant trees was approximately 20 m. Regeneration of desirable species was mostly restricted to smaller (< 30 cm tall) height classes (Raymond et al., 2018). It included balsam fir (50 690 seedlings·ha^{−1}, of which 95% were < 30 cm), yellow birch (35 270 seedlings·ha^{−1}, of which 98% were < 30 cm), red maple (6220 seedlings·ha^{−1}, of which 80% were < 30 cm), and sugar maple (6110 seedlings·ha^{−1}, of which 86% were < 30 cm), with sparse red spruce (2150 seedlings·ha^{−1}, of which 77% were < 30 cm). Non-commercial species, namely mountain maple (*Acer spicatum* Lamarck), striped maple (*Acer pensylvanicum* Linné), and American mountain-ash (*Sorbus americana* Marshall) were dominant in larger height classes (12 130 seedlings·ha^{−1}, of which 58% were > 30 cm).

The second silvicultural experiment of the SSAM project (Silvicultural Systems Adapted to Mixedwood forests), established on this site in 2007–2008 (Raymond et al., 2018), was used for the present study. The experimental design is a split-plot with five complete randomized blocks (repetitions, based on hardwood/conifer proportions), each comprising four cutting treatments, for a total of 20 experimental units (EU, 80 m × 80 m). Cutting treatments include an uncut control (26 m²·ha^{−1} of BA, with mean natural canopy gap areas of ≤ 100 m² each) and three hybrid single-tree and group-selection cuttings corresponding to a light cut (20% removal; residual BA [RBA]: 21 m²·ha^{−1}; mean harvest gap area: 220 m²), a moderate cut (31% removal; RBA: 18 m²·ha^{−1}; mean gap area: 270 m²), and a heavy cut (42% removal; RBA: 15 m²·ha^{−1}; mean gap area: 340 m²). Each EU was split in two (40 m × 80 m) sub-treatments: natural regeneration and enrichment planting with red spruce.

Trees were marked to be harvested as single and small groups, according to priorities based on vigor (Boulet, 2005), species (priority given to trees with DBH > 20 cm for balsam fir and > 32 cm for paper birch) and quality (defective trees removed first). Harvest was conducted during fall 2008. Trees were felled using a chainsaw, then skidded with a cable Timberjack 240E, which disrupted the forest floor within harvested areas (passive scarification), but avoided damaging the roots of living trees and spruce saplings. Hardwoods were delimbed on site and skidded tree-length to the landings, whereas softwoods were skidded full-tree to the landings (Raymond et al., 2018).

2.2. Plant material and planting conditions

Large-sized (46.6 cm tall with root collar diameter of 5.1 mm) red spruce seedlings were container-grown (IPL® [Saint-Damien, QC, Canada] 25–310 containers: 25 cells, 310 cm³ each) in a forest nursery (Reboisement Mauricie Inc., Saint-Étienne-des-Grès, QC, lat. 46° 26' N, long. 72° 47' W, alt. 61 m) for two years following standard practices for seedling production in the province of Quebec (Tétreault et al., 1990). Containers were filled with a peat moss and vermiculite substrate (4:1, v/v). The first year, seedlings grew in a tunnel covered with a polyethylene film filtering 50% of sunlight. The second year, the film was removed to allow seedlings to acclimatize to fully exposed microclimatic conditions. They were shipped to the planting site the following year.

Seedlings were manually planted in the spring of 2009, only within the harvest gaps created by the different cuts, with a spacing of 3 m. Seedlings were also planted systematically in the control plots (spacing of 6 m), for comparison. A total of 1750 seedlings were planted (~88 per EU). They were planted in mineral soil or in a mix of mineral and organic soils, in order to standardize planting substrate conditions, to respect government standards (MFFP, 2016) and to enhance growth performance (e.g., Thiffault et al., 2013). In July 2012, about three years after planting, a mechanical release treatment was applied in half of each planted plot (40 m × 40 m subplots) to limit the abundant regrowth of non-commercial competing species. For this, competing vegetation was removed 1.5 m around and 2 m above each planted seedling using a brush saw.

2.3. Light availability

Initial (2009, post-cut) summer light availability was characterized in each EU using hemispherical photographs (Coolpix 4500 digital camera and FC-E88-mm fisheye lens, Nikon, Tokyo, Japan) taken 1.3 m above the center of each vegetation plot (9 microplots per EU, Raymond et al., 2018) under overcast sky. All photographs were analyzed with gap light analyzer (GLA) software to calculate % of total transmitted light (Frazer et al., 1999).

2.4. Ecophysiological variables monitoring

Three of the five experimental blocks were randomly selected to collect ecophysiological data. Measurements were taken on sunny days from 2010 to 2015 (two dates per year [mid-July and mid-August], except in 2010 and 2015 [only date in mid-August]) on three randomly selected seedlings per subplot, for a total of 72 observations (3 blocks × 4 cutting treatments × 2 release treatments × 3 seedlings) at each sampling date. Measurements were made on the same seedlings during the entire study (repeated measures).

Gas exchange measurements were carried out at ambient CO₂ concentration (~400 ppm), using fully-developed current-year foliage on a lateral branch that was randomly selected in the upper half of the seedling shoot. Transpiration (E), stomatal conductance (g), light-saturated photosynthesis ($A_{\max}$) and water use efficiency ($WUE = A_{\max} \times E^{-1}$) were measured between 9:00 A.M. and 4:30 P.M. (solar time) using a portable photosynthesis system (LI-6400, LI-COR BioSciences, Lincoln, NE, USA) equipped with an opaque conifer chamber (6400–22L, LI-COR). Light saturation (photosynthetic photon flux density of ~1 000 $\mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$, ~50% of full sunlight) in the chamber was provided by an adapted LED lamp unit (6400–18 RGB light source, LI-COR). $A_{\max}$ measurement allows the estimation of photosynthetic capacity (Man and Loeffers, 1997) and the verification of whether physiological acclimation occurs (Mitchell, 2001). Air temperature and relative humidity inside the chamber were maintained around 24 °C and 50%. After gas exchange measurements, the section of the branch inserted in the chamber was used to determine daytime shoot water potential (Ψ_d) with a pressure chamber (Model 600, PMS

Instruments Co., Corvallis, OR, USA). It was then enclosed in a hermetic plastic bag, placed in a portable cooler, and promptly transported to a cold room (–10 °C) in prevision of subsequent analyses. In the laboratory, needles of this branch section were scanned, and their projected area was determined using imaging software (WinSEEDLE, Régent Instruments Inc., Québec, QC, Canada). Needles were then dried at 65 °C for 48 h and weighed to calculate the specific leaf area ($SLA = [\text{needle area}] \times [\text{needle dry weight}]^{-1}$). SLA measurement allows the detection of foliage acclimation and measures its effectiveness to intercept the available light (Mailly and Kimmins, 1997).

2.5. Nutrient status assessment

Concentrations of main mineral nutrients (N, P, K, Ca, Mg) were determined from the needle sample used to calculate SLA . Laboratory analyses were performed on 100 mg of ground dry needles digested with concentrated sulfuric acid in the presence of selenium and hydrogen peroxide at 370 °C for 1 h (Walinga et al., 1995). Kjeldahl N was determined by calorimetry (Quikchem 8000, Lachat Instruments, Milwaukee, WI, USA). Foliage concentrations of P, K, Ca and Mg were determined by plasma atomic emission spectrometry (ICAP 61E, Thermo Jarrel Ash Instruments, Franklin, MA, USA).

2.6. Survival, growth and morphological measurements

Seedling survival was recorded annually in fall. Planted spruces were classified as dead if their needles were completely yellow, orange, or dry. Seedling height (H) and root collar diameter (D , hereafter referred to as “diameter”) were measured at planting (spring 2009) and then annually (in fall) until 2016. H/D ratio was calculated to assess the degree of seedling stability (Navratil, 1996). Height growth (HG , length of the leader), lateral growth (LG , using only one random branch per whorl, adjacent to the leader measured for HG ; Prévost et al., 2016), diameter growth (DG , annual increment), the number of branches composing the annual whorl (NBW) and the number of annual internodal branches (NIB , appearing on the previous-year's leader) were measured annually. NBW and NIB were determined as indicators of light conditions, morphological acclimation, and vigor (Ruel et al., 2000, Prévost et al., 2016). The HG/LG ratio was calculated to assess the degree of apical control (Greis and Kellomäki, 1981, Ruel et al., 2000).

2.7. Data analyses

Statistical analyses were performed using the MIXED procedure of the SAS software (version 9.4, SAS Institute, Cary, NC, USA). For ecophysiological and nutritional variables, data of the two sampling dates of each year were pooled for each planted seedling, and analysis was performed on the calculated mean. Physiological, morphological and growth data were submitted to analyses of variance for repeated measures (ANOVAR) with a first-order autoregressive structure (AR[1]). To avoid confounding effects, data were separated in two periods: before release (2010–2011) and after release (2012–2016), and an ANOVA was made for each period. Cutting treatment, release treatment, year and their interactions were considered as fixed effects, whereas block was considered as a random effect. Homogeneity of the variance and normality of residuals were verified graphically. When necessary, mathematical transformations were used (square root for HG , natural logarithm for D), but after analysis, data were back-transformed to their original scale for presentation. Planned contrasts were used to separate treatment effects into single degrees of freedom. All tests were considered significant at $P \leq 0.05$.

Table 1

Analyses of variance and associated probabilities ($P > F$) for the effects of cutting treatments on survival, height (H), diameter (D) and H/D ratio of planted red spruce seedlings in 2016, eight growing seasons after planting.

Source of variation	Ndf ^a	Ddf ^b	Survival	Height	Diameter	H/D ratio
Cutting treatment (C) ^c	3	16.3	< 0.001	< 0.001	< 0.001	0.026
Control vs All cuts ^d	1	16.0	< 0.001	< 0.001	< 0.001	0.009
Light cut vs Moderate cut	1	15.8	0.023	0.313	0.089	0.109
Moderate cut vs Heavy cut	1	16.7	0.356	0.009	0.345	0.966
Release (R)	1	16.1	0.013	< 0.001	< 0.001	< 0.001
C × R	3	16.0	0.025	0.007	0.041	0.037
(Control vs All cuts) × R	1	15.8	0.358	0.003	< 0.001	0.425
(Light cut vs Moderate cut) × R	1	15.6	0.006	0.801	0.396	0.017
(Moderate cut vs Heavy cut) × R	1	16.5	0.505	0.046	0.754	0.735

Significant P -values are presented in **boldface type**.

^a Numerator degrees of freedom.

^b Denominator degrees of freedom (example provided for Survival). To simplify table presentation, Ddf of the other variables, some non-relevant contrasts, as well as random effects are not shown.

^c Control = 0% removal of merchantable basal area (BA, area of all trees > 9 cm diameter at 1.3 m above the ground level), BA of 26 m²·ha⁻¹, with mean natural canopy gap areas of ≤ 100 m² each. Light cut = 20% BA removal, residual BA (RBA) of 21 m²·ha⁻¹, with mean harvest gap areas of 220 m² each (the smallest gaps). Moderate cut = 31% BA removal, RBA of 18 m²·ha⁻¹, with mean harvest gap areas of 270 m² each. Heavy cut = 42% BA removal, RBA of 15 m²·ha⁻¹, with mean harvest gap areas of 340 m² each (the largest gaps) (Raymond et al., 2018). In the control, spruce seedlings were planted systematically (spacing of 6 m), while in the cuts, seedlings were planted only in harvest gaps (spacing of 3 m).

^d All cuts = Light cut + Moderate cut + Heavy cut.

3. Results

3.1. Light availability

Initial (2009) summer light availability was significantly different among cutting treatments. The % of total transmitted light varied according to a consistent gradient (heavy cut > moderate cut > light cut > uncut control), with mean values of 33.2%, 19.8%, 17.6%, and 13.3% of full light at 1.3 m, respectively ($P = 0.002$; Raymond et al., 2018). When compared with the control, the gain of light availability was greater in the heavy cut (+20%) than in moderate and light cuts (+4–7%).

3.2. Seedling survival

Survival of planted seedlings decreased slightly and gradually over time. By 2016, it reached 84% in the heavy and moderate cuts, and 77% in the light cut ($P \leq 0.023$; Table 1, Fig. 1a). In the control, survival dropped rapidly during the first two years after planting, reaching 49% in 2010, then decreased more gradually to 37% by 2016. Survival in 2016 was related to the interaction between cutting and release treatments ($C \times R$, $P = 0.025$; Table 1). Release enhanced survival only in the control (+12%) and in the light cut (+21%, $P = 0.006$; Fig. 1b). Thus, released seedlings in the light cut maintained a survival rate (~85%) comparable to that of observed in moderate and heavy cuts.

3.3. Ecophysiological variables

Before release (2010–2011), Ψ_d was 27% lower in the heavy cut than in the moderate cut (−1.12 MPa vs −0.88 MPa, $P = 0.009$; Table 2, Fig. 2a). After release (2012–2015), there was no effect of release on Ψ_d (Fig. 2b), but this variable increased slightly over time, from −0.99 MPa in 2012 to −0.85 MPa in 2014 ($P = 0.003$; not shown). No significant effect of cutting or release treatment was detected for E and g . After release, E decreased over time from 1.8 mmol·m⁻²·s⁻¹ in 2012 to 0.8 mmol·m⁻²·s⁻¹ in 2015 ($P < 0.001$; Table 2, not shown). Before release, g decreased from 2010 to 2011 (0.16 to 0.13 mol·m⁻²·s⁻¹, $P < 0.001$; not shown). After release, it increased up to 0.17 mol·m⁻²·s⁻¹ in 2014, and then dropped again to 0.13 mol·m⁻²·s⁻¹ in 2015 ($P = 0.001$).

Generally, A_{max} increased with cutting intensity ($P \leq 0.049$; Fig. 2c–d). After release, A_{max} was affected by the $C \times R$ interaction

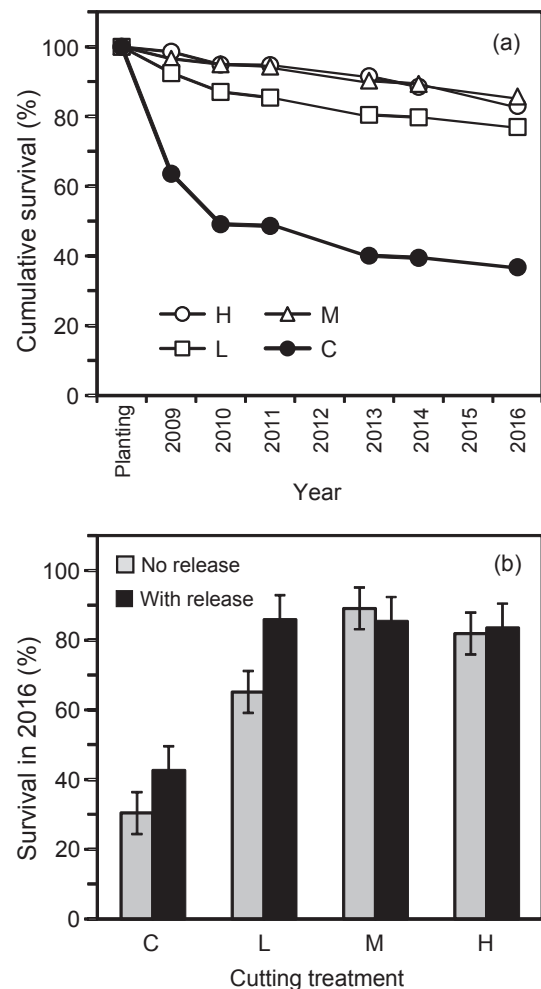


Fig. 1. (a) Effects of cutting treatments (C: control, L: light cut, M: moderate cut, H: heavy cut) on cumulative survival of planted red spruce seedlings. Maximum standard error (MSE) was $\pm 4\%$, $\pm 5\%$, $\pm 2\%$, and $\pm 2\%$, respectively for C, L, M, and H. (b) Effects of cutting and release treatments on seedling survival eight growing seasons after planting. Cutting treatments are described in Table 1 (footnote c). Error bars represent the MSE.

Table 2

Analyses of variance for repeated measures and associated probabilities ($P > F$) for the effects of cutting treatment on daytime shoot water potential (Ψ_d), transpiration (E), stomatal conductance (g), light-saturated photosynthesis ($A_{\max}$), water use efficiency (WUE), and specific leaf area (SLA) of planted red spruce seedlings.

Source of variation	Ndf ^a	Ddf ^b	Ψ_d	E	g	$A_{\max}$	WUE	SLA
<i>Before release (2010–2011)</i>								
Cutting treatment (C)	3	4.9	0.027	0.702	0.840	< 0.001	0.036	0.002
Control vs All cuts	1	4.9	0.125	0.980	0.596	< 0.001	0.039	< 0.001
Light cut vs Moderate cut	1	4.9	0.305	0.399	0.896	0.003	0.018	0.712
Moderate cut vs Heavy cut	1	4.9	0.009	0.324	0.730	0.008	0.216	0.043
Year (Y)	1	1.0	0.183	0.117	< 0.001	0.159	0.413	0.535
C × Y	3	1.0	0.996	0.783	0.177	0.819	0.940	0.782
<i>After release (2012–2015)</i>								
C	3	8.3	0.780	0.691	0.519	0.015	0.078	0.002
Control vs All cuts	1	8.5	0.904	0.768	0.158	0.011	0.126	< 0.001
Light cut vs Moderate cut	1	8.4	0.331	0.333	0.764	0.548	0.409	0.849
Moderate cut vs Heavy cut	1	8.4	0.688	0.939	0.824	0.049	0.165	0.045
Release (R)	1	22.4	0.113	0.476	0.370	< 0.001	0.003	< 0.001
C × R	3	22.3	0.917	0.351	0.157	0.006	0.001	0.221
(Control vs All cuts) × R	1	22.5	0.678	0.337	0.680	0.024	< 0.001	0.113
(Light cut vs Moderate cut) × R	1	22.6	0.579	0.144	0.787	0.523	0.276	0.710
(Moderate cut vs Heavy cut) × R	1	22.6	0.624	0.468	0.099	0.005	0.334	0.390
Year (Y) ^c	3	42.8	0.003	< 0.001	0.001	< 0.001	< 0.001	< 0.001
C × Y	9	42.8	0.813	0.919	0.107	0.336	0.485	0.210
R × Y	3	34.6	0.454	0.092	0.134	0.416	0.075	0.202
C × R × Y	9	34.6	0.326	0.882	0.597	0.915	0.872	0.099

Cutting treatments are described in Table 1 (footnote c). Significant P -values are presented in **boldface type**.

^a Numerator degrees of freedom.

^b Denominator degrees of freedom (example provided for Ψ_d). To simplify table presentation, Ddf of the other ecophysiological variables, some non-relevant contrasts, as well as random effects are not shown.

^c Exceptionally, Ndf = 2 for Ψ_d (after release) because this variable was measured only during the first three years after release (2012–2014).

($P = 0.006$; Table 2). Released seedlings had significantly greater $A_{\max}$ only in the heavy cut (+27%, $P = 0.005$; Fig. 2d). All treatments combined, $A_{\max}$ changed over time, from $6.4 \mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$ in 2012 to $12.7 \mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$ in 2013 and to $7.5 \mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$ in 2015 ($P < 0.001$; not shown). Before release, WUE was higher in the moderate cut (+33%) than in the light cut ($P = 0.018$; Table 2, Fig. 2e). After release, WUE was affected by the $C \times R$ interaction ($P = 0.001$). Released seedlings had a greater WUE in cuts compared to the control (+20%, $P < 0.001$; Fig. 2f). All treatments combined, WUE increased over time, from $4.2 \mu\text{mol}\cdot\text{mmol}^{-1}$ in 2012 to $9.2 \mu\text{mol}\cdot\text{mmol}^{-1}$ in 2015 ($P < 0.001$; not shown).

In general, SLA decreased as cutting intensity increased ($P \leq 0.045$; Table 2, Fig. 2g–h). Release reduced SLA to a similar extent in all cuts (−9%, $P < 0.001$; Fig. 2h). After release, SLA decreased over time, from $59.2 \text{ cm}^2\cdot\text{g}^{-1}$ in 2012 to $49.2 \text{ cm}^2\cdot\text{g}^{-1}$ in 2015 ($P < 0.001$; not shown).

3.4. Mineral nutrient concentrations

In general, foliar concentrations of main mineral nutrients in planted seedlings varied among years ($P \leq 0.039$; Table 3), but no effect of cutting treatment was detected, except for N before release ($18 \text{ mg}\cdot\text{g}^{-1}$ in the control vs $15.4 \text{ mg}\cdot\text{g}^{-1}$ in harvest gaps, $P = 0.040$; not shown). All treatments combined, released seedlings had slightly lower concentrations of N (13.7 vs $14.6 \text{ mg}\cdot\text{g}^{-1}$, $P = 0.017$; not shown), P (1.6 vs $1.7 \text{ mg}\cdot\text{g}^{-1}$, $P = 0.003$) and K (6.8 vs $7.8 \text{ mg}\cdot\text{g}^{-1}$, $P = 0.002$), but a slightly higher concentration of Ca (1.6 vs $1.3 \text{ mg}\cdot\text{g}^{-1}$, $P < 0.001$).

3.5. Growth and morphological variables

Generally, HG and DG were greater in cuts ($9.5 \text{ cm}\cdot\text{year}^{-1}$ and $1.9 \text{ mm}\cdot\text{year}^{-1}$, respectively) than in the control ($2.5 \text{ cm}\cdot\text{year}^{-1}$ and $0.4 \text{ mm}\cdot\text{year}^{-1}$, $P < 0.001$; Table 4, Figs. 3a, c and 4a–d). After release (2012–2016), HG and DG were related to the $C \times R$ interaction ($P < 0.001$ and 0.036 , respectively). Released seedlings had greater

HG and DG only in cuts ($P \leq 0.001$; Table 4, Fig. 4b and d). Consequently, height (2016) increased to a similar degree in light and moderate cuts (+24%), and even more in the heavy cut (+36%, $P = 0.046$; Table 1, Fig. 3b), compared to the control. A similar increase in D (2016) was observed in all cuts (+30%, Fig. 3d) compared to the control.

Before release, seedling H/D and HG/LG ratios were higher in cuts than in the control (both $P = 0.009$; Tables 1 and 4, Fig. 4e for HG/LG). After release, H/D and HG/LG ratios were affected by the $C \times R$ interaction ($P \leq 0.048$; Tables 1 and 4). Released seedlings had smaller H/D ratios only in the control (−8%) and in the light cut (−11%, $P = 0.017$; Fig. 3e). Consequently, released seedlings in the light cut had the same H/D ratio (~63) as in moderate and heavy cuts. Released seedlings had a larger HG/LG ratio only in the heavy cut (+37%, $P = 0.006$; Table 4, Fig. 4f).

In general, the NBW was greater in cuts (3.4 branches) than in the control (1.4 branches, $P \leq 0.013$, Table 4; Fig. 4g–h). Overall, released seedlings also had greater NBW (+29%, $P < 0.001$; Fig. 4h). Before release, the NIB was higher in light and moderate cuts (2.1 branches) than in the control (0.9 branch), and even more in the heavy cut (3.8 branches, $P = 0.043$; Fig. 4i). After release, NIB was affected by the $C \times R$ interaction ($P = 0.028$), because release enhanced NIB only in cuts (+85%, $P = 0.047$; Fig. 4j).

4. Discussion

4.1. Cutting treatment effects on microclimate and seedling performance

As expected, survival and morphophysiological performance of planted red spruce seedlings were much better in hybrid single-tree and group-selection cutting treatments, where 24% to 60% more light was available than in the uncut control. This confirms our first prediction and corroborates results of Dumais et al (2019) indicating that red spruce seedlings, despite their high tolerance to shade, perform better when planted in more open conditions. According to observations made in the control treatment, it seems that traditional (full sunlight-

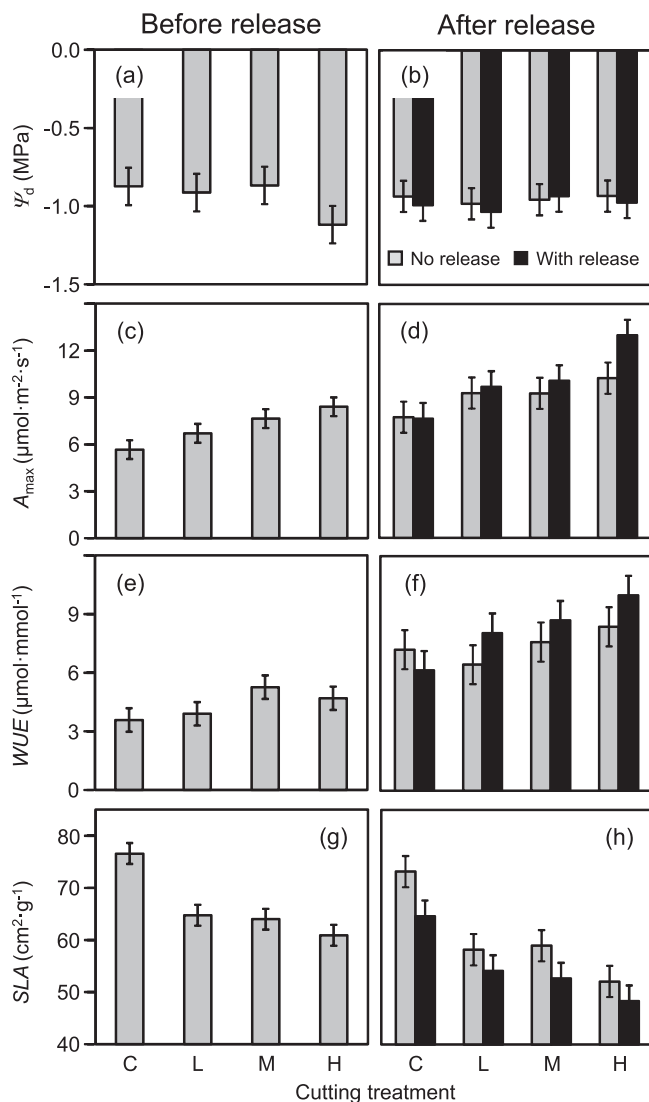


Fig. 2. Effects of cutting treatments (C: control, L: light cut, M: moderate cut, H: heavy cut) before (2010–2011) and after (2012–2015) release on (a–b) daytime shoot water potential (Ψ_d), (c–d) light-saturated photosynthesis (A_{max}), (e–f) water use efficiency (WUE), and (g–h) specific leaf area (SLA) of planted red spruce seedlings. In all cases, error bars represent the maximum standard error.

acclimated) nursery seedlings of red spruce are not suited for understory planting, and that this activity should be performed under a cover that is at least partially open. Low survival rate observed in the uncut control (< 40% by 2016, Fig. 1a) indicates that planted seedlings have difficulty establishing and acclimating to lower light conditions simultaneously. Immediately after planting, access to soil water is generally restricted by poor root-soil contact (Margolis and Brand, 1990). These deficiencies may be overcome by root growth, which depends largely on current photosynthesis (Burdett, 1990), which in turn was lower in the control than in harvest gaps (Fig. 2c–d). Nevertheless, our ecophysiological data indicate that surviving seedlings in the control (located within or at the border of small natural canopy gaps) could establish and acclimate to these understory conditions. For example, the greater SLA in the control (Fig. 2g–h) indicates a morphological acclimation of seedling foliage, likely in response to a low-light environment (Greenwood et al., 2009). However, acclimation to shade in spruce species is known to occur at the expense of the photosynthetic capacity (e.g., Grassi and Bagnaresi, 2001, Dumais and Prévost, 2008), as we observed in the present study (Fig. 2c–d). Indeed, morphophysiological performance of planted seedlings in the uncut control of the

present study was comparable to that of small oppressed natural red spruce regeneration growing under closed canopies (Dumais and Prévost, 2008, 2014).

In support of our second prediction, results show that morphophysiological performance of planted seedlings was better in the most intense treatment (heavy cut with harvest gaps averaging 340 m² in area). Increased HG and DG in these gaps (Figs. 3 and 4) can be explained by greater photosynthesis (Fig. 2) and more internodal branches (Fig. 4i–j). Both are positively correlated to light availability (Ruel et al., 2000, Prévost et al., 2016) and allow total carbon gain to increase (Givnish, 1988). However, the level of transmitted light (33% on average) remains lower than the 50% threshold established for optimal physiology and growth of red spruce (Blum, 1990, Seymour, 1995). This suggests that, as expected, seedling growth was not optimal in these harvest gaps. Indeed, red spruce HG could average up to 20 cm·year⁻¹ to 25 cm·year⁻¹ during the first five years in larger openings (Dumais and Prévost, 2008, Dumais et al., 2019), while our best growth reached 15 cm·year⁻¹ in the present study. The complex light availability dynamics in harvest gaps of the present study (e.g., strong border effect, seasonal variability) probably did not create adequate conditions for the optimal ecophysiology and development of planted nursery seedlings.

Temperature stress is known to strongly affect the ecophysiology of young red spruce trees (Vann et al., 1994, Alexander et al., 1995, Day, 2000). When compared with an uncut control, soil temperature increased by 5 °C in 4900 m² patch clearcuts (Prévost, 2008). As well, increases of 1–2 °C and 3–4 °C were reported by Prévost and Raymond (2012), respectively in 300–700 m² circular harvest gaps and in 10 000 m² clearcuts. Although we did not present temperature data in the present study, the fact that seedlings showed no signs of physiological stress in the 340 m² gaps of the heavy cut suggests that these were small enough to maintain the microclimate (Fig. 2).

4.2. Release effects on seedling performance

In support of our third prediction, the release treatment had a positive effect on survival and morphophysiological performance of planted red spruce seedlings, but not significantly in all cutting treatments. Indeed, release enhanced survival only in the uncut control and in the smallest harvest gaps (220 m², Fig. 1b). This indicates that under such closed canopies, the shade from surrounding vegetation (often abundant in yellow birch – conifer stands; Prévost, 2008, Prévost and Charette, 2015, 2019) can be an additional obstacle to the establishment of planted nursery red spruce seedlings. The observed morphological adjustments (SLA, NBW, NIB; Figs. 2 and 4) show that light availability improved substantially after release, since red spruce requires a large change in light conditions to express its morphological plasticity (Dumais and Prévost, 2008, 2014). Morphological acclimation was probably the main cause of the 12% survival gain observed in the released control plots (Fig. 1b), because physiological data remained unchanged (Fig. 2). Delayed physiological response after microenvironmental changes is well known for red spruce (Dumais and Prévost, 2007). However, in the short term, acclimation of SLA and an increased NBW and NIB can enhance photosynthetic efficiency and total carbon gain (Givnish, 1988, Evans and Poorter, 2001). In the smallest gaps, the increase of WUE (Fig. 2f) may also have contributed to enhance seedling survival after release.

Short-term light enhancement provided by the release treatment was not sufficient to allow rapid gains in HG and DG in the uncut control (Fig. 4). This is probably because the main limiting factor was light interception by the forest canopy (Dumais et al., 2019). It is also possible that the small carbon gain was first allocated to root growth (Burdett, 1990). By causing a decrease in H/D ratio (Fig. 3e), the release treatment could have maintained some seedling stability and vigor (Cole and Newton, 1987) in low-light environments where crown shape was flatter (according to the lower HG/LG ratio in the control;

Table 3

Analyses of variance for repeated measures and associated probabilities ($P > F$) for the effects of cutting treatment on nutrient concentrations in the current-year foliage of planted red spruce seedlings: nitrogen (N), phosphorus (P), potassium (K), calcium (Ca), and magnesium (Mg).

Source of variation	Ndf ^a	Ddf ^b	N	P	K	Ca	Mg
<i>Before release (2010–2011)</i>							
Cutting treatment (C)	3	12.0	0.168	0.933	0.626	0.537	0.426
Control vs All cuts	1	12.0	0.040	0.983	0.459	0.608	0.747
Light cut vs Moderate cut	1	12.0	0.663	0.740	0.413	0.942	0.632
Moderate cut vs Heavy cut	1	12.0	0.924	0.892	0.340	0.347	0.268
Year (Y)	1	16.0	0.959	0.003	0.275	0.038	0.021
C × Y	3	16.0	0.974	0.761	0.726	0.378	0.601
<i>After release (2012–2015)</i>							
C	3	12.0	0.789	0.623	0.717	0.290	0.218
Control vs All cuts	1	11.8	0.381	0.464	0.650	0.116	0.331
Light cut vs Moderate cut	1	12.0	0.734	0.305	0.333	0.995	0.298
Moderate cut vs Heavy cut	1	11.9	0.983	0.527	0.462	0.321	0.366
Release (R)	1	20.9	0.017	0.003	0.002	< 0.001	0.794
C × R	3	20.8	0.914	0.697	0.308	0.454	0.302
(Control vs All cuts) × R	1	21.9	0.512	0.715	0.551	0.669	0.622
(Light cut vs Moderate cut) × R	1	19.6	0.908	0.370	0.740	0.139	0.741
(Moderate cut vs Heavy cut) × R	1	20.8	0.838	0.976	0.201	0.312	0.111
Year (Y)	3	46.9	< 0.001	0.006	0.039	0.307	< 0.001
C × Y	9	47.0	0.111	0.492	0.142	0.211	0.971
R × Y	3	73.2	0.226	0.366	0.618	0.976	0.712
C × R × Y	9	73.0	0.519	0.238	0.955	0.591	0.493

Cutting treatments are described in Table 1 (footnote c). Significant P -values are presented in **boldface type**.

^a Numerator degrees of freedom.

^b Denominator degrees of freedom (example provided for N). To simplify table presentation, Ddf of the other growth variables, some non-relevant contrasts, as well as random effects are not shown.

Fig. 4e–f). By contrast, release generally increased morphophysiological performance in all hybrid single-tree and group-selection cutting treatments, with greatest gains in the heavy cut with the largest gaps for many variables including photosynthesis (Figs. 2–4). These results concord with those of Dumais and Prévost (2008, 2014) and suggest that growth performance of red spruce regeneration depends mainly on carbon gain brought about by enhanced photosynthesis in response to increased light availability.

4.3. Management implications

Red spruce decline represents an important biodiversity issue in managed temperate mixedwood forests of northeastern North America (Dumais and Prévost, 2007), particularly in uneven-aged yellow birch-conifer stands growing on rich sites (Duchesne and Prévost, 2013). Species diversity, stand structure, and gap dynamics greatly contribute to the ecological value of this forest type. However, maintaining these important attributes through conventional silviculture is

Table 4

Analyses of variance for repeated measures and associated probabilities ($P > F$) for the effects of cutting treatment on height growth (HG), diameter growth (DG), ratio of HG over lateral growth (HG/LG ratio), number of branches composing the annual whorl (NBW), and number of annual internodal branches (NIB) of planted red spruce seedlings.

Source of variation	Ndf ^a	Ddf ^b	HG	DG	HG/LG ratio	NBW	NIB
<i>Before release (2010–2011)</i>							
Cutting treatment (C)	3	12.0	< 0.001	< 0.001	0.034	0.040	0.026
Control vs All cuts	1	12.0	< 0.001	< 0.001	0.009	0.013	0.024
Light cut vs Moderate cut	1	12.0	0.213	0.194	0.271	0.148	0.567
Moderate cut vs Heavy cut	1	12.0	0.363	0.268	0.401	0.721	0.043
Year (Y)	1	16.0	< 0.001	0.420	0.072	0.269	0.488
C × Y	3	16.0	0.645	0.850	0.110	0.986	0.447
<i>After release (2012–2016)</i>							
C	3	12.0	0.002	0.003	0.022	0.016	0.011
Control vs All cuts	1	12.0	0.001	< 0.001	0.007	0.004	0.025
Light cut vs Moderate cut	1	11.9	0.691	0.621	0.294	0.190	0.582
Moderate cut vs Heavy cut	1	11.9	0.046	0.207	0.385	0.463	0.019
Release (R)	1	20.9	< 0.001	< 0.001	0.110	< 0.001	0.005
C × R	3	20.8	< 0.001	0.036	0.048	0.631	0.028
(Control vs All cuts) × R	1	21.8	0.001	0.020	0.708	0.389	0.047
(Light cut vs Moderate cut) × R	1	19.4	0.826	0.808	0.652	0.793	0.638
(Moderate cut vs Heavy cut) × R	1	20.8	0.001	0.160	0.006	0.387	0.251
Year (Y)	4	46.7	< 0.001	< 0.001	0.130	0.385	0.460
C × Y	12	47.0	0.220	0.100	0.103	0.527	0.098
R × Y	4	73.2	0.130	0.922	0.602	0.592	0.560
C × R × Y	12	73.1	0.853	0.312	0.413	0.414	0.672

Cutting treatments are described in Table 1 (footnote c). Significant P -values are presented in **boldface type**.

^a Numerator degrees of freedom.

^b Denominator degrees of freedom (example provided for HG). To simplify table presentation, Ddf of the other growth variables, some non-relevant contrasts, as well as random effects are not shown.

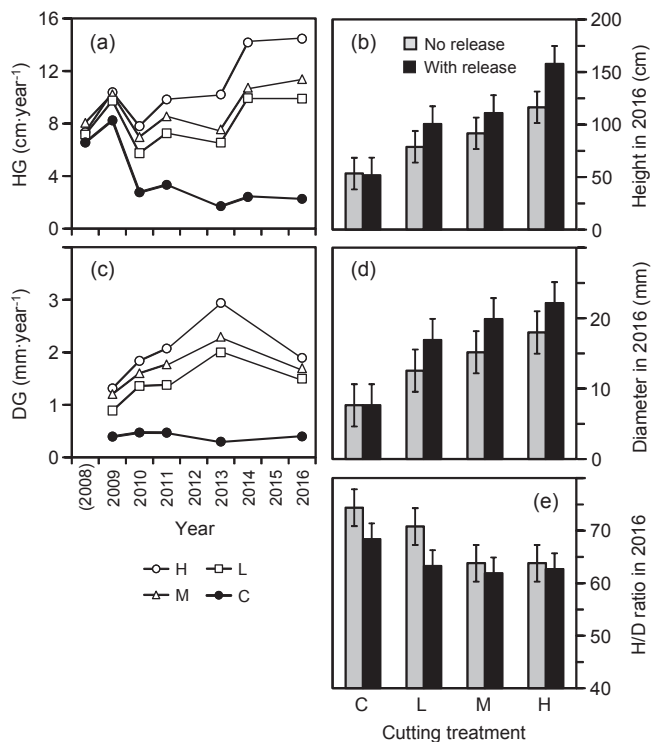


Fig. 3. (a) Effects of cutting treatments (C: control, L: light cut, M: moderate cut, H: heavy cut) on height growth (HG) of planted red spruce seedlings over time. Maximum standard error (MSE) was ± 0.5 $\text{cm}\cdot\text{year}^{-1}$, ± 2.1 $\text{cm}\cdot\text{year}^{-1}$, ± 1.3 $\text{cm}\cdot\text{year}^{-1}$, and ± 2.6 $\text{cm}\cdot\text{year}^{-1}$, respectively, for C, L, M, and H. (b) Effects of cutting and release treatments on height, eight growing seasons after planting (2016). Error bars represent the MSE. (c) Effects of cutting treatment on diameter growth (DG). MSE was ± 0.1 $\text{mm}\cdot\text{year}^{-1}$, ± 0.4 $\text{mm}\cdot\text{year}^{-1}$, ± 0.2 $\text{mm}\cdot\text{year}^{-1}$, and ± 0.4 $\text{mm}\cdot\text{year}^{-1}$, respectively for C, L, M, and H. (d) Effects of cutting and release treatments on diameter in 2016. Error bars represent the MSE. (e) Effects of cutting and release treatments on H/D ratio in 2016. Error bars represent the MSE.

challenging. Even if planting red spruce seedlings in small clearcut patches (4900–12 100 m^2) could give interesting results (Dumais et al., 2019), this approach is difficult to reconcile with the natural dynamics of yellow birch–conifer stands (Kneeshaw and Prévost, 2007, Duchesne and Prévost, 2013). According to results of the present study, enrichment planting in harvest gaps of partially cut stands could be seen as a means to maintain the red spruce component and integrity in these mixedwood stands without interfering with other biodiversity and socioeconomic issues, such as regenerating high-value yellow birch (Prévost and Charette, 2015). The present study also showed that mechanical release a few years after planting improves the survival, establishment, physiology and growth of planted red spruce seedlings, as was previously reported for white spruce in other enrichment planting experiments (Hébert et al., 2013, Dumais and Prévost, 2019, Prévost and Charette, 2019). However, managing competing vegetation while promoting regeneration and optimal growth of desirable tree species in yellow birch–conifer stands remain a great challenge, especially in an ecosystem-based management context. Finally, because red spruce is known to have development patterns which are mainly associated with multiple-tree gap release events, a gradual opening of the forest cover could be a future means to allow the recruitment of planted seedlings in the upper canopy (Fraver and White, 2005, Dumais and Prévost, 2007, Kneeshaw and Prévost, 2007, Duchesne and Prévost, 2013).

5. Synthesis and conclusion

The present study tested enrichment planting of red spruce in a

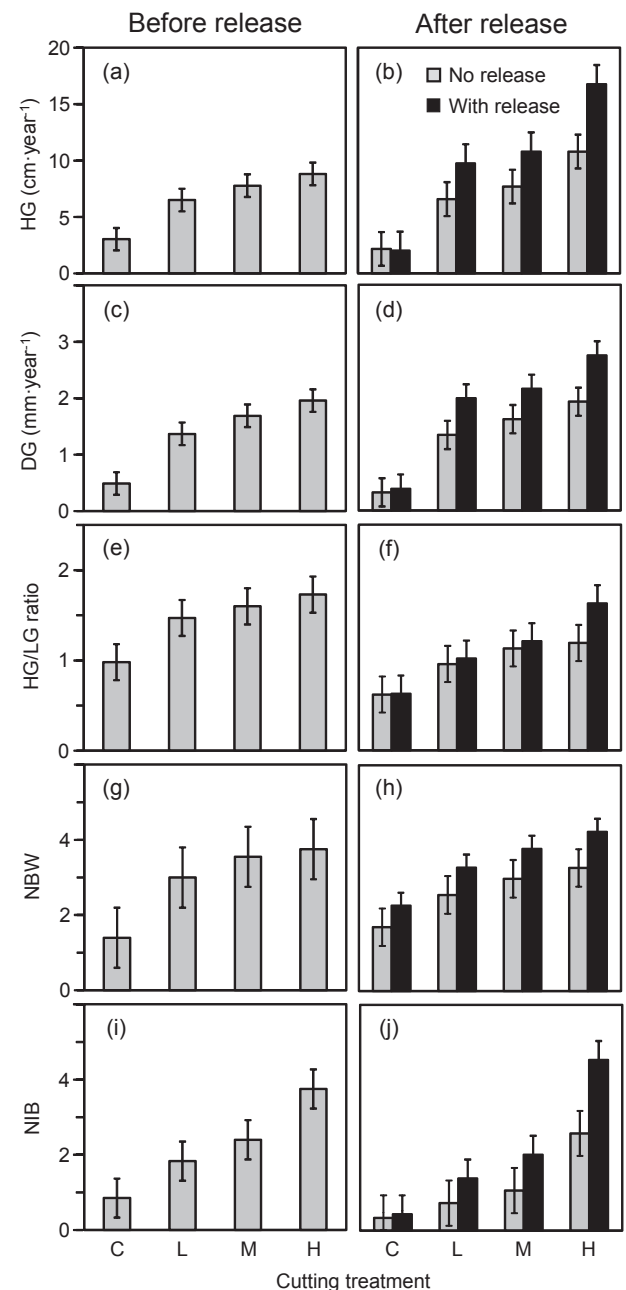


Fig. 4. Effects of cutting treatment (C: control, L: light cut, M: moderate cut, H: heavy cut) before (2010–2011) and after (2012–2016) release on: (a–b) height growth (HG), (c–d) diameter growth (DG), (e–f) ratio of HG over lateral growth (HG/LG ratio), (g–h) number of branches composing the annual whorl (NBW), and (i–j) number of annual internodal branches (NIB) of planted red spruce seedlings. In all cases, error bars represent the maximum error standard.

managed yellow birch–conifer stand. We aimed to characterize morphophysiology, acclimation, and growth of nursery seedlings planted under intact forest canopy (uncut control) and in a harvest gradient with intermediate gap sizes (averaging 220–340 m^2 in area) created by three treatments of hybrid single-tree and group-selection cutting. Results after eight years showed that seedling survival was equally high in harvest gaps and greater than in the control. Ecophysiological and growth monitoring indicated that planted red spruce seedlings performed best in the largest gaps. Release treatment enhanced survival in the control and in the smallest gaps, and generally increased physiological and growth performances (which, however, remained sub-optimal) in harvest gaps, particularly in the largest ones. Even if red

spruce is very shade-tolerant, low survival and performance observed in the control suggest that seedlings grown in open conditions with traditional forest nursery methods are not adapted to planting under a closed forest canopy. Based on our results, enrichment planting in intermediate-size harvest gaps (of about 340 m² in area) of partially cut stands (of about 40% in BA) combined to mechanical release could be a good strategy to maintain or reintroduce red spruce and promote its growth, while regenerating yellow birch. In the future, periodical measurements of planted spruce seedlings would be necessary to provide long-term information on aspects such as survival, health condition, development, and recruitment.

CRedit authorship contribution statement

Daniel Dumais: Conceptualization, Methodology, Formal analysis, Writing - original draft, Visualization. **Patricia Raymond:** Conceptualization, Methodology. **Marcel Prévost:** Conceptualization, Methodology.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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